Diogo Ferreira

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PROFESSIONAL EXPERIENCE

CICANT (Universidade Lusófona) Lisbon, Portugal

Full-Stack AI Engineer February 2025 –Present

LusoChat (*AI-powered course assistant using retrieval-augmented generation****)***

- Developed an intelligent chatbot system combining FastAPI and Gemini (Generative AI) with a RAG (retrieval-augmented generation) pipeline to answer student queries about university courses.
- Built a vector database from SQL dumps using embedding models (e.g., SentenceTransformers) to semantically match user queries to relevant academic programs.
- Implemented dual-stage inference: first detecting user intent, then using Gemini to shortlist relevant courses before selecting the best match via vector similarity.
- Enabled natural language queries on curriculum, ECTS credits, tuition fees, and more by grounding large language model responses in institution-specific data.
- Architected the backend in Python (FastAPI) and the frontend in React, ensuring seamless, real-time user interaction with AI-powered responses.

Tools: Python, FastAPI, React, Gemini API, SentenceTransformers, SQLite, Git, RESTful APIs

Full-Stack AI Engineer February 2023 –Present

Cyanotypes (*Matchmaking platform for the creative sector* ***)***

- As part of the EU project CYANOTYPES, I developed a smart matchmaking platform that connects educators and creative industry professionals. I used artificial intelligence and natural language processing (including Google's Gemini API) to automatically match people based on their profiles and interests.
- Led the platform's design and built the back end using Laravel, with FastAPI and Docker handling the matchmaking logic.
- Implemented NLP-based matching using text embeddings and vector databases to connect users based on profile similarity.
- Collaborated with technical and academic teams in the Innovation Department and supported digital research infrastructure and dissemination.

Tools: Laravel, JavaScript, Docker, Fast Api, Python, RESTful APIs, Google Gemini Api, Git, PHP, MySQL

Web Developer February 2024 –Present

CIEL (*Plugin-based system to automate web deployment across multiple higher education institutions*)

- Designed and developed a modular plugin with a scheduled cronjob to integrate internal academic data via API across Lusophone Education institutions (e.g., ISLA Santarém, ISMAT).
- Automated the creation and deployment of institutional websites, reducing setup time to under 10 minutes per instance.
- Enabled seamless synchronization between backend academic systems and institutional web frontends, ensuring data consistency and scalability.
- Built continuous content update mechanisms through scheduled tasks, keeping institutional websites up-to-date in real time.

Tools: Joomla Framework, PHP, RESTful APIs, Git, Cron

Web Developer February 2024 –January 2025

Kuadro (*Room scheduling platform for academic institutions****)***

- Developed a Laravel-based scheduling system to enable teachers and staff to reserve rooms, view existing bookings, and avoid scheduling conflicts.
- Integrated an interactive calendar interface using a JavaScript library (e.g., FullCalendar.js) to provide real-time, visual room availability.
- Implemented authentication and role-based access controls to tailor features for different user types (e.g., admin, teacher).
- Created intuitive weekly/monthly views for seamless classroom management and transparent collaboration among academic staff.

Tools: Laravel, PHP, MySQL, JavaScript, FullCalendar.js, Blade Templates, RESTful APIs, Git

Web Developer February 2022 –June 2023

Nervo Photobooks *(Digital platform for cataloging and promoting Portuguese photobooks)*

- Developed and maintained a Laravel-based web application for cataloging and showcasing Portuguese photobooks.
- Implemented advanced search and filtering functionalities to enhance user navigation through the photobook collection.
- Collaborated with a multidisciplinary team to ensure the platform met the needs of researchers, artists, and the general public.
- Contributed to a project recognized by national media for its role in preserving and promoting Portuguese photographic heritage. (Público Journal Article)

Tools: Laravel, PHP, MySQL, JavaScript, Blade Templates, Git, RESTful APIs

EDUCATION

Universidade Lusófona Lisbon, Portugal

Bachelor's degree in Information Management September 2021 – July 2025

- Relevant Coursework: Project Based Learning (PBL), Object-oriented programming, Business Intelligence

Universidade Lusófona Lisbon, Portugal

Master's degree in Data Science September 2025 – July 2027

- Relevant Coursework: NLP(Natural Language Processing), RAG and Machine Learning

SKILLS

- **Programming Languages:** Python, PHP (Laravel), JavaScript, HTML, CSS, SQL
- **Frameworks & Tools:** Laravel, FastAPI, Docker, Composer, npm
- **APIs & Libraries:** Google Gemini API, Natural Language Processing (NLP), Sentence Transformers, REST API
- **Databases:** MySQL, MariaDB
- **Data & AI:** Text Embeddings, NLP Pipelines, Vector Similarity Search
- **Version Control:** Docker

PROFESSIONAL ACHIEVEMENTS

International Presentations Brussels, Belgium

March 2024

Represented Universidade Lusófona at conferences and events in Belgium, where I was responsible for presenting and demonstrating internally developed technology platforms. These presentations included technical explanations of AI-powered matchmaking systems and web modules designed to support scientifc research.

Technical Leadership in European Contexts Matera, Italy

April 2025

Acted as the lead technical representative in international settings, recognized for my clarity in communicating complex concepts and ability to tailor presentations to multidisciplinary audiences, including researchers, academic staff, and institutional decision-makers.

Applied AI in Creative Skills Week Prague, Czechia

September 2025

Represented Universidade Lusófona at the Creative Skills Week, presenting the AI-powered matchmaking platform developed under the Cyanotypes project. Demonstrated how NLP and vector-based similarity can connect creative professionals and educators across Europe. Engaged with international stakeholders from academia, industry, and policy-making, showcasing the role of AI in supporting creative sector innovation.